

Sandy Aoun

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HIGHER EDUCATION

Doctoral Student in Computer Science (Dec. 2016 - Feb. 2017)

University of Rennes I, MATISSE Doctoral School, IRISA, Lannion, France

Research topic: Recording Scripts Optimization for the Expressive Reading (Speech Synthesis) of Audiobooks

M.Sc. in Computer Science (Oct. 2015 - Sept. 2016)

Lebanese University, Faculty of Sciences I, Hadath, Lebanon

University of Toulouse III, Faculty of Science and Engineering, IRIT, Toulouse, France

Double degree program, Track: IRDBM (Information Retrieval, Database, and Multimedia)

Master's thesis topic: Optimal Constitution of the Speech Corpus for the Speech Synthesis of Audiobooks

Completed with Honors - *Mention Assez Bien*, Rank: 1/10

First year of graduate studies in Computer Science (Oct. 2014 - June 2015)

Lebanese University, Faculty of Sciences II, Fanar, Lebanon

Completed with High Honors - *Mention Bien*, Rank: 3/19

B.Sc. in Computer Science (Oct. 2009 - June 2014)

Lebanese University, Faculty of Sciences II, Fanar, Lebanon

Graduated with Honors - *Mention Assez Bien*

RESEARCH EXPERIENCE

Research Assistant (Aug. 2022 - July 2026)

Institution: University of Graz, Department of Digital Humanities, Graz, Austria

Research topic: Information Extraction from Late Medieval Charters

Supervisor: Prof. Dr. Georg Vogeler

Disciplines: Natural Language Processing, Digital Humanities, Lexical and Relational Semantics, Machine Learning

Research Assistant (Apr. 2022 - June 2022)

Institution: American University of Beirut, Faculty of Engineering and Architecture, Beirut, Lebanon

Research topic: Arabic Graph-Based Named Entity Linking

Supervisor: Dr. Fadi A. Zaraket

Scientific disciplines: Natural Language Processing, Lexical and Relational Semantics, Knowledge Graphs

Research Assistant (Sept. 2019 - Nov. 2020)

Institution: American University of Beirut, Faculty of Arts and Sciences, Beirut, Lebanon

Research topic: Automatic Speech Recognition of Arabic Speech using Sequence-to-Sequence Models

Supervisor: Prof. Dr. Wassim El-Hajj

Scientific disciplines: Natural Language Processing, Machine Learning, Speech Processing

Research Internship of the Master's Program in Computer Science (Mar. 2016 - Sept. 2016)

Institution: University of Rennes I, ENSSAT, IRISA, Team EXPRESSION, Lannion, France

Research topic: Optimal Constitution of the Speech Corpus for the Speech Synthesis of Audiobooks

Supervisors: Dr. Jonathan Chevelu, Dr. Ali Choumane, and Prof. Dr. Damien Lolive

Scientific disciplines: Natural Language Processing, Combinatorial Optimization, Speech Processing

ENGINEERING EXPERIENCE

Natural Language Processing Engineer (June 2021 - Sept. 2021)

Institution: University of California at Berkeley, College of Letters & Science, United States

Topic: Construction of a Biological Database from 'Flora in the Eastern Mediterranean' Reference Books

Collaborator: Maryam Sedaghatpour (Ph.D. student in Integrative Biology)

Scientific disciplines: Natural Language Processing, Information Extraction, Data Cleansing

Final Project of the First Year of Graduate Studies in Computer Science (Mar. 2015 - May 2015)

Institution: Lebanese University, Faculty of Sciences II, Fanar, Lebanon

Topic: Conception and Implementation of a JSON to XML Compiler

Mentor: Prof. Dr. Kablan Barbar

Scientific disciplines: Formal Language Theory, Compiler Design, Compiler Construction

TEACHING EXPERIENCE

Natural Language Processing Tutor (July 2024)

The Department of Digital Humanities at the University of Graz organized a summer school titled: “Computational Language Technologies for Medievalists” in July 2024. The school’s objective is to provide students with a solid Natural Language Processing foundation which would enable them to apply NLP tasks and techniques to historical medieval manuscripts. I was tasked with equipping the students with essential knowledge on the information extraction NLP task and its sub-challenges, along with covering the process of creating Named Entity Recognition datasets, and showcasing my scholarly work in this area.

Computer Science Tutor (April 2021 - Sept. 2021)

I privately tutored programming and computer science-related concepts. Programming-related concepts: variables (built-in types and associated built-in functions); operators; basic programming syntax; basic manipulation of command line; control statements; looping/iterating; built-in functions; the “main” function; user-defined functions; external modules; specific modules. Computer science-related concepts: algorithm; pseudo-code; programming language theory; data structure; functional programming. Self-made appropriate programming examples and exercises are used to illustrate the concept(s) in question. [Programming language used: *Python*.]

SCHOLARLY OUTPUT

PUBLICATIONS

Franziska Decker, Sandy Aoun, Giuseppe Consolo. “**From Documents to Data: Digital Technologies in the Study of Notarial Charters**”. *XIV convegno annuale dell’Associazione per l’Informatica Umanistica e la Cultura Digitale (AIUCD 2025)*, Verona, Italy, June 11-13, 2025.

Sandy Aoun, Varvara Arzt, Daniel Luger, Georg Vogeler. “**Information Extraction from German Medieval Charters Abstracts**”. *19th Annual Conference of the Alliance of Digital Humanities Organizations (DH 2024)*, Washington, D.C., United States, August 6-10, 2024. [I also prepared and performed a 10-minute Oral Presentation.]

Florian Atzenhofer-Baumgartner, Daniel Luger, Tamás Kovács, Johannes Laroche, Angelos Nicolaou, Franziska Decker, Nicolas Renet, Sandy Aoun, Niklas Tscherne, Georg Vogeler. “**Formulaic Language in Diplomats: Investigating Formulas as Charter Type Discriminators**”. *Conference on Formulaic Language in Historical Research and Data Extraction*, Amsterdam, The Netherlands, February 7-9, 2024.

Andreas Habring, Angelos Nicolaou, Daniel Luger, Florian Atzenhofer-Baumgartner, Florian Lamminger, Franziska Decker, Sandy Aoun, Tamás Kovács, Georg Vogeler, Martin Holler. “**Probabilistic Modeling of Chronological Dates to Serve Machines and Scholars**”. *18th Annual Conference of the Alliance of Digital Humanities Organizations (DH 2023)*, Graz, Austria, July 10-14, 2023.

Tamás Kovács, Sandy Aoun, Georg Vogeler, Angelos Nicolaou, Daniel Luger, Florian Atzenhofer-Baumgartner, Florian Lamminger, Franziska Decker. “**Few Shot Classification for Labeling of Medieval and Early Modern Charter Texts**”. *18th Annual Conference of the Alliance of Digital Humanities Organizations (DH 2023)*, Graz, Austria, July 10-14, 2023.

Daniel Luger, Angelos Nicolaou, Franziska Decker, Florian Atzenhofer-Baumgartner, Florian Lamminger, Georg Vogeler, Sandy Aoun, Tamás Kovács. “**Digital Contributions to a 300 Years Old Methodology: Diplomats & DH**”. *18th Annual Conference of the Alliance of Digital Humanities Organizations (DH 2023)*, Graz, Austria, July 10-14, 2023.

Georg Vogeler, Daniel Luger, Angelos Nicolaou, Tamás Kovács, Florian Atzenhofer-Baumgartner, Florian Lamminger, Sandy Aoun, Franziska Decker. “**Building a Virtual Research Environment to Move from Digital to Distant Diplomats (ERC Project DiDip)**”. *9. Tagung des Verbands Digital Humanities im deutschsprachigen Raum (DHd 2023)*, Belval, Luxembourg and Trier, Germany, March 13-17, 2023.

Georg Vogeler, Angelos Nicolaou, Daniel Luger, Tamás Kovács, Florian Atzenhofer-Baumgartner, Sandy Aoun, Franziska Decker. “**Computational Methods in Studying Late Medieval Charters**”. *Third Conference on Computational Humanities Research (CHR 2022)*, Antwerp, Belgium, December 12-14, 2022.

OTHER SCHOLARLY MANUSCRIPTS

Sandy Aoun and Wassim El-Hajj. “**Automatic Speech Recognition of Arabic Speech Using Sequence-to-Sequence Models**”. Submitted to the Grant Research Program which is jointly supported by the American University of Beirut and the National Council for Scientific Research (Lebanon). 2019/2020 Academic Year.

Sandy Aoun. “**Optimal Constitution of the Speech Corpus for the Speech Synthesis of Audiobooks**”. Lebanese University and University of Toulouse III - M.Sc. Thesis in Computer Science. September 2016. Written in French. [Thesis defense: I also prepared and performed a *20-minute Oral Presentation*.]

Sandy Aoun. “**Conception and Implementation of a JSON to XML Compiler**”. Lebanese University - Graduate Research Project in Computer Science. May 2015. Written in French. [Project defense: I also prepared and performed a *30-minute Oral Presentation*.]

RESEARCH SOFTWARE

Constructing Bilad al-Sham Flora Database: I implemented software programs which transform unstructured factual text input into a valuable biological database. In essence, useful/specific information is extracted from encyclopedia-like PDF files covering flora in the Eastern Mediterranean. The extracted data is consecutively refined into a standardized database. 2021. [Programming language used: *Python*.]

Building End-to-End ASR Dataset: I carried out an experiment which addresses building datasets suitable for training end-to-end automatic speech recognition (ASR) systems of spoken Arabic dialects. Our proposed automatic dataset collection method consists of firstly crawling YouTube videos whose Arabic closed captions are provided by the channel owner (the most frequent words in Arabic tweets are used as search keywords); then secondly passing the videos and their associated captions through several filtering heuristics which ensure reaching a satisfactory outcome. I also developed a program which aims to assess the effectiveness of our approach by generating relevant statistics. 2020. [Technologies used: *Python, Bash, YouTube Search API, SoX*.]

Packaging MGB-2 Dataset: I implemented software programs which process the MGB-2 dataset [Ali+16] in order to ultimately convert it into a form readable by the pipeline of the high-performance speech recognition framework Wav2letter++ [Pra+19]. 2019. [Technologies used: *Python, SoX, Wav2letter++* [Pra+19], *Docker*.]

Optimal Constitution of TTS Speech Corpus: I carried out an experiment which aspires to optimize the process of constructing the speech corpus of unit selection text-to-speech (TTS) systems. In this context, I implemented a greedy algorithm (spitting strategy) to bring into view the trade-off between the amount of text to be recorded and the quality of obtained (synthesized) speech signals. The implementation is based on our formal theoretical analysis which essentially profits from concepts related to the following domains/sub-domains: Set Cover Problem; Approximation Algorithms; and Linear Algebra. 2016. [Technologies used: *Python, IRISA TTS System* [Ala+16], *ROOTS* [CLL14].]

Objective Evaluation of Speech Signals: I implemented a software which measures the objective distance between natural and synthesized speech signals. In our case, the objective distance corresponds to the normalized Dynamic Time Warping cost which is computed on the Euclidean Distance between the Mel-Generalized Cepstral sequences of the signals. 2016. [Technologies used: *Python, SPTK* (Speech Signal Processing Toolkit), *SoX* (Sound eXchange).]

JSON to XML Compiler: I implemented a compiler which translates a JSON-formatted document into an interchangeable XML-formatted document. The implementation is based on my theoretical analysis which amounted to firstly defining a formal grammar as well as formulating a lexical and syntactic analysis of the syntax of JSON, then subsequently devising a semantic analysis by coming up with a suitable attribute grammar. 2015. [Programming language used: *C++*.]

PROFESSIONAL ACTIVITIES

ACADEMIC SERVICE

Journal Reviewer

ACM Transactions on Asian and Low-Resource Language Information Processing (2020)

Conference Reviewer

4th Workshop on Open-Source Arabic Corpora and Processing Tools (of LREC 2020)

Annual Conference of the Alliance of Digital Humanities Organizations (DH 2024, DH 2025, DH 2026)

Summer School Organizational Support

I contributed to the organization of the “Computational Language Technologies for Medievalists” Summer School (University of Graz, July 2024)

CONFERENCE/SEMINAR ATTENDANCE

I attended the 2016 **Annual Seminar** of the **EXPRESSION Team** (University of South Brittany, June 2016)

I attended (and volunteered at) the **Digital Diplomats Conference 2022** (University of Graz, September 2022)

I attended (and volunteered at) the **Digital Humanities Conference 2023** (University of Graz, July 2023)

I attended the **Digital Humanities Conference 2024** (George Mason University, August 2024)

AWARDS

I received a **Grant of 850 Euros** from the Alliance of Digital Humanities Organizations to **Attend the Digital Humanities Conference 2024**. This award is aimed at students and early career scholars, with the selection process focusing on the scholarly merit of their submission.

SKILLS

TECHNICAL SKILLS

Programming languages: Python, C, C++, Java, Lisp, SQL

Miscellany: HTML, CSS, LaTeX, Linux, Docker, HPC clusters, pandas, spaCy, scikit-learn

NATURAL LANGUAGES PROFICIENCY

English: Full professional proficiency (IELTS Academic test: CEFR level C1)

Arabic: Native or bilingual proficiency

French: Professional working proficiency

German: Elementary proficiency

BIBLIOGRAPHY

- [Ala+16] Pierre Alain et al. “The IRISA text-to-speech system for the Blizzard Challenge 2016”. In: *Blizzard Challenge 2016 workshop*. 2016.
- [Ali+16] Ahmed Ali et al. “The MGB-2 challenge: Arabic multi-dialect broadcast media recognition”. In: *IEEE Spoken Language Technology Workshop (SLT)*. IEEE. 2016, pp. 279–284.
- [CLL14] Jonathan Chevelu, Gwénolé Lecorvé, and Damien Lolive. “ROOTS: a toolkit for easy, fast and consistent processing of large sequential annotated data collections.” In: *International Conference on Language Resources and Evaluation (LREC)*. 2014, pp. 619–626.
- [Pra+19] Vineel Pratap et al. “Wav2letter++: A fast open-source speech recognition system”. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2019, pp. 6460–6464.

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